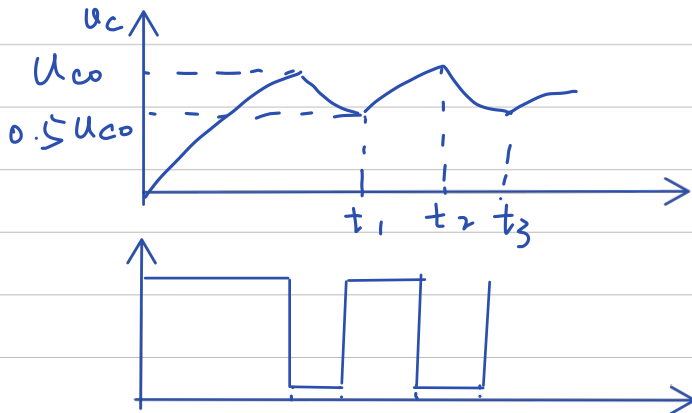


6.2.5.

$$f_0 = \frac{1}{0.693} \frac{1}{C(R_1 + 2R_2)} = 0.05 \mu F.$$

6.2.6.



$$u_c(t^+) = 0.5 U_{co} \quad u_c(\infty) = U_{DD}$$

$$\therefore u_c = U_{DD} + (0.5 U_{co} - U_{DD}) e^{\frac{t-t_1}{(R_1+R_2)C}}$$

$$t = t_2 \text{ 时}$$

$$U_{co} = U_{DD} + (0.5 U_{co} - U_{DD}) e^{\frac{t_2-t_1}{(R_1+R_2)C}}$$

$$T_1 = (R_1 + R_2) C \ln \frac{0.5 U_{co} - U_{DD}}{U_{co} - U_{DD}}$$

$$t_2 \text{ 时放电} \quad u_c(t_2^+) = U_{co}$$

$$u_c(\infty) = 0$$

$$u_c = U_{co} e^{-\frac{t-t_2}{R_2 C}}$$

$$t = t_3 \quad 0.5 U_{co} = U_{co} e^{\frac{t_3-t_2}{R_2 C}}$$

$$\therefore T_2 = R_2 C \ln 2$$

$$T = T_1 + T_2 = (R_1 + R_2) C \ln \frac{0.5 U_{co} - U_{DD}}{U_{co} - U_{DD}} + R_2 \ln 2$$